CNSCC.405, Lab 3

A. Okumura/Hata Model

Okumura模型是通过大量的实验获得的，用于计算移动的通信系统的天线高度和覆盖区域。它是最常用的路径损耗模型之一

The Okumura model has been obtained through extensive experiments to compute the antenna height and coverage area for mobile communication systems. It is one of the most frequently adopted path loss models that can predict path loss in an urban area. This particular model mainly covers the typical mobile communication system characteristics with a frequency band of 500–1500MHz, cell radius of 1–100 km, and an antenna height of 30 m to 1000 m.

The path loss at distance d in the Okumura model is given as

$$PL_{Ok}(d)[dB] = PL_F + A_{MU}(f,d) - G_{Rx} - G_{Tx} + G_{AREA} \quad (1)$$

where $A_{MU}(f,d)$ is the medium attenuation factor at frequency f, G_{Rx} and G_{Tx} are the antenna gains of Rx and Tx antennas, respectively, and G_{AREA} is the gain for the propagation environment in the specific area. Note that the antenna gains, G_{Rx} and G_{Tx}, are merely a function of the antenna height, without other factors taken into account like an antenna pattern. Meanwhile, $A_{MU}(f,d)$ and G_{AREA} can be referred to by the graphs that have been obtained empirically from actual measurements by Okumura

The Okumura model has been extended to cover the various propagation environments, including urban, suburban, and open area, which is now known as the Hata model. In fact, the Hata model is currently the most popular path loss model. For the height of transmit antenna, h_{TX}[m], and the carrier frequency of f_c[MHz], the path loss at distance d [m] in an urban area is given by the Hata model as

$$PL_{Hata,U}(d)[dB] = 69.55 + 26.16\log f_c - 13.82\log h_{TX} - C_{RX} + (44.9 - 6.55\log h_{TX})\log d$$

where C_{RX} is the correlation coefficient of the receive antenna, which depends on the size of coverage. For small to medium-sized coverage, C_{RX} is given as

$$C_{Rx} = 0.8 + (1.1\log f_c - 0.7)h_{Rx} - 1.56\log f_c$$

where h_{RX} [m] is the height of transmit antenna. For large-sized coverage, C_{RX} depends on the range of the carrier frequency, for example,

Ref: MIMO-OFDM wireless communications with MATLAB. Cho, Y. S., Kim, J., Yang, W. Y., & Kang, C. G. John Wiley & Sons, 2010.

$$C_{RX} = \begin{cases} 8.29(\log(1.54h_{RX}))^2 - 1.1 & \text{if } 150\,\text{MHz} \le f_c \le 200\,\text{MHz} \\ 3.2(\log(11.75h_{RX}))^2 - 4.97 & \text{if } 200\,\text{MHz} \le f_c \le 1500\,\text{MHz} \end{cases}$$

Meanwhile, the path loss at distance d in suburban and open areas are respectively given by the Hata model as

$$PL_{Hata,SU}(d)[dB] = PL_{Hata,U}(d) - 2\left(\log\frac{f_c}{28}\right)^2 - 5.4$$

and

$$PL_{Hata,O}(d)[dB] = PL_{Hata,U}(d) - 4.78(\log f_c)^2 + 18.33\log f_c - 40.97$$

Figure 1 presents the path loss graphs for the three different environments – urban, suburban, and open areas. It is clear that the urban area gives the most significant path loss as compared to the other areas,

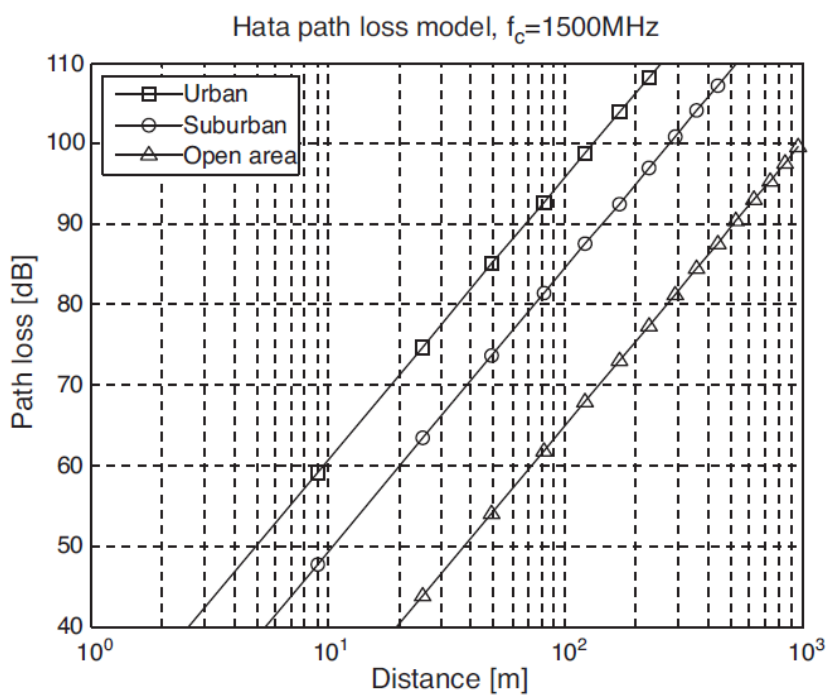

Figure 1: Hata path loss model

Tasks: 1. 绘制三种不同环境下的路径损耗图--如上所述的城市、郊区和开阔地区。（如果需要，可以做任何实际的假设）。

Plot the path loss graphs for the three different environments – urban, suburban, and open areas as described above. (Make any realistic assumptions if required).

Ref: MIMO-OFDM wireless communications with MATLAB. Cho, Y. S., Kim, J., Yang, W. Y., & Kang, C. G. John Wiley & Sons, 2010.

B. IEEE 802.16d Model

IEEE 802.16d model is based on the log-normal shadowing path loss model. There are three different types of models (Type A, B, and C), depending on the density of obstruction between the transmitter and receiver (in terms of tree densities) in a macro-cell suburban area. Table 1 describes these three different types of models in which ART and BRT stand for Above-Roof-Top and Below-Roof-Top. The IEEE 802.16d path loss model is given as

$$PL_{802.16}(d)[dB] = PL_F(d_0) + 10\gamma \log_{10}\left(\frac{d}{d_0}\right) + C_f + C_{RX} \quad \text{for} \quad d > d_0$$

In Equation , $d_0 = 100\,m$ and $\gamma = a - bh_{Tx} + c/h_{TX}$ where a, b, and c are constants that vary with the types of channel models as given in Table 1.3, and h_{TX} is the height of transmit antenna (typically, ranged from 10 m to 80 m). Furthermore, C_f is the correlation coefficient for the carrier frequency f_c [MHz], which is given as

$$C_f = 6\log_{10}(f_c/2000)$$

Table 1: Types of IEEE 802.16d path loss models.

Type	Description
A	Macro-cell suburban, ART to BRT for hilly terrain with moderate-to-heavy tree densities
B	Macro-cell suburban, ART to BRT for intermediate path loss condition
C	Macro-cell suburban, ART to BRT for flat terrain with light tree densities

Table 2: Parameters for IEEE 802.16d type A, B, and C models.

Parameter	Type A	Type B	Type C
a	4.6	4	3.6
b	0.0075	0.0065	0.005
c	12.6	17.1	20

Meanwhile, C_{RX} is the correlation coefficient for the receive antenna, given as

$$C_{RX} = \begin{cases} -10.8\log_{10}(h_{RX}/2) & \text{for Type A and B} \\ -20\log_{10}(h_{RX}/2) & \text{for Type C} \end{cases}$$

Ref: MIMO-OFDM wireless communications with MATLAB. Cho, Y. S., Kim, J., Yang, W. Y., & Kang, C. G. John Wiley & Sons, 2010.

or

$$C_{RX} = \begin{cases} -10\log_{10}(h_{RX}/3) & \text{for } h_{RX} \le 3m \\ -20\log_{10}(h_{RX}/3) & \text{for } h_{RX} > 3m \end{cases}$$

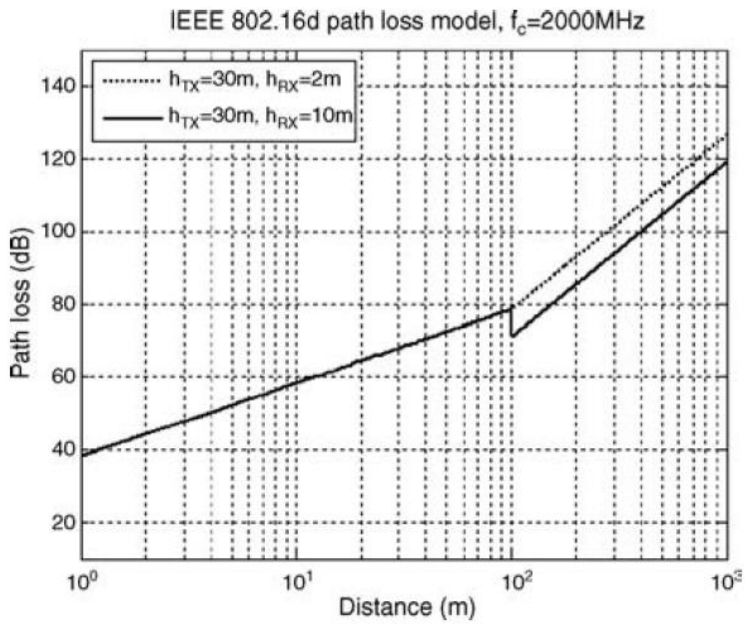

Figure 2: IEEE 802.16d path loss model

Tasks: 2.

Plot the path loss graphs the path loss by the IEEE 802.16d model as described above. (Make any realistic assumptions if required). 根据上述IEEE 802.16d模型绘制路径损耗图。（如果需要，请进行任何实际假设）。

Tasks: 3.

Write a summary of what you have learned, observed, things went wrong (if in your experiment, you have not had the predicted result), and how you can overcome the similar problem in future

写一个总结，说明你所学到的，观察到的，出了什么问题（如果在你的实验中，你没有得到预期的结果），以及你将来如何克服类似的问题

Ref: MIMO-OFDM wireless communications with MATLAB. Cho, Y. S., Kim, J., Yang, W. Y., & Kang, C. G. John Wiley & Sons, 2010.